

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Duration: 60 Minutes

Total Marks: 20

Code of Conduct

I 192372280 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ramakrishna

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty

Total Marks Awarded:

Assessment Tasks

Concept Mapping Tasks

- Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical flow.
1. Concept Map 1: Cloud Computing Architecture: Include front-end, back-end, client devices, data center, and distributed servers.
 2. Concept Map 2: Characteristics of Cloud Computing: Connect on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.
 3. Concept Map 3: Deployment Models: Map public cloud, private cloud, hybrid cloud, and community cloud with usage scenarios.
 4. Concept Map 4: Cloud Benefits vs Challenges: Show advantages (cost efficiency, scalability, flexibility) and limitations (security, downtime, vendor lock-in).

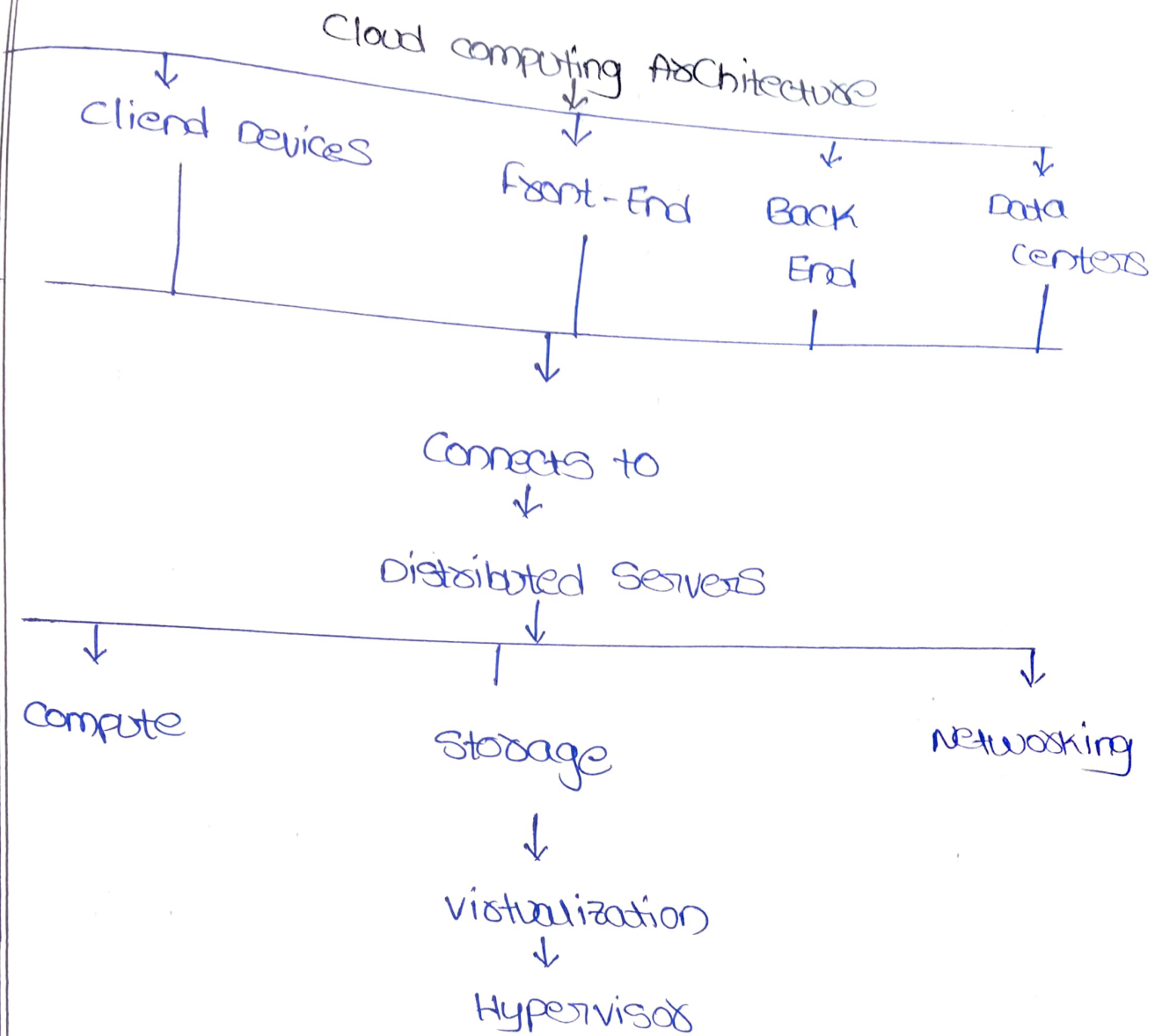
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Cloud computing Architecture

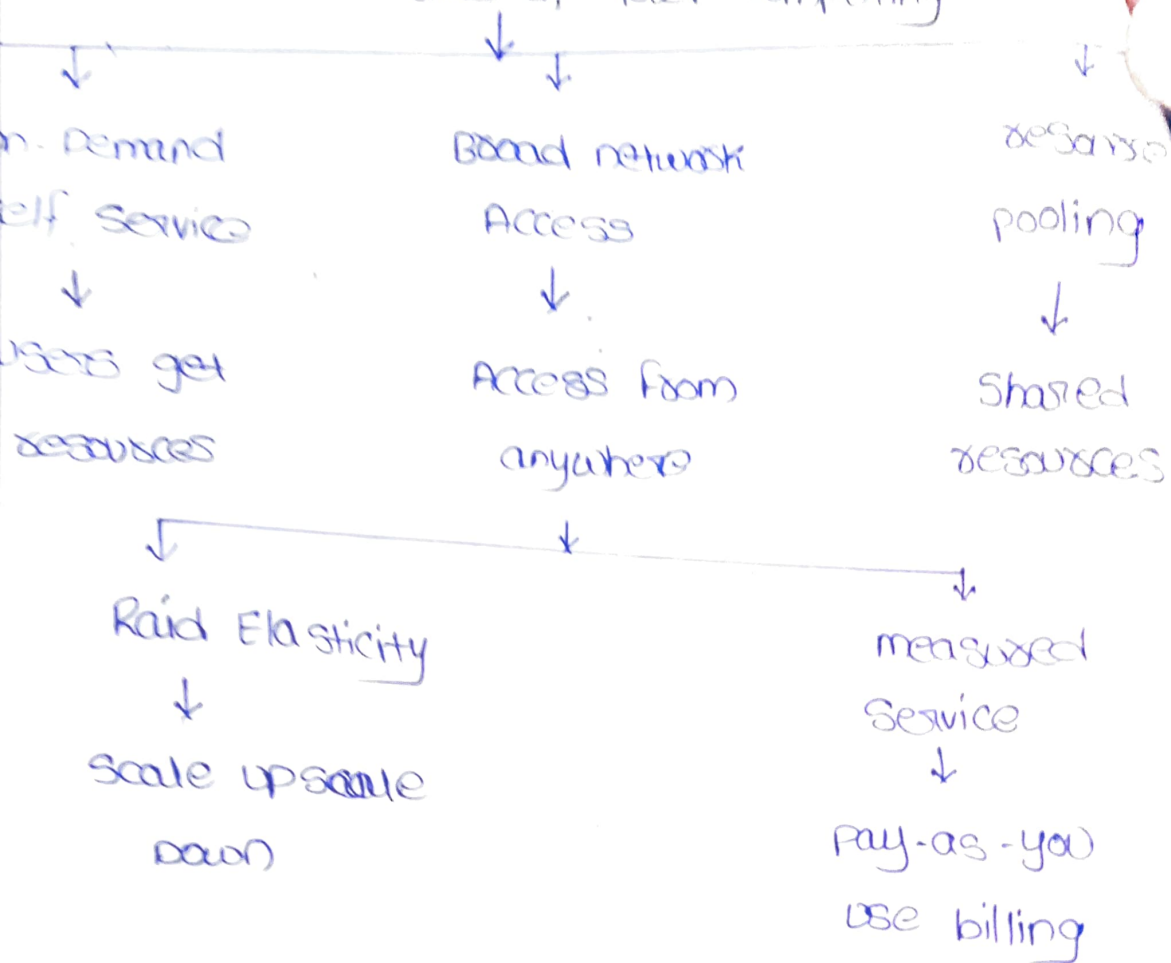


Points:-

- * Cloud architecture has Front-End and Back-End
- * Front-End includes client devices
- * Back-End includes servers, database, applications, and storage
- * Data Centers store and manage data
- * Both parts communicate through the Internet

Characteristics of cloud computing

Characteristics of cloud computing



points:-

on-demand self-service :- Users access resources anytime

Broad network access :- services are available over the internet

Resource pooling :- Resources are shared among multiple users

Measured service :- Users pay only for the resources they use.

Concept map: cloud deployment models.

Information:-

Deployment models describes how cloud services are made available to users

main nodes:-

cloud deployment models

- * public cloud
- * private cloud
- * Hybrid cloud
- * Community cloud

public cloud:-

1. used by general public
2. Ex:- web hosting, startups

private cloud:-

1. used by single organisation
2. Ex: Banks, hospitals

Hybrid cloud:-

1. Combines public and private clouds

Ex:- companies storing sensitive data privately

4. Concept map: Cloud Benefits vs Challenges

Information:-

cloud computing offers many advantages but also introduces certain risks and limitations

main nodes:-

Benefits:-

1. cost Efficiency
2. scalability
3. flexibility
4. High Availability
5. easy maintenance

Challenges:-

1. security Risks
2. downtime
3. vendor lock in
4. privacy concerns
5. Internet dependency.